The documentation and process conversion measures necessary to comply with this revision shall be completed by 1 August 1995.

INCH-POUND

MIL-S-19500/374C
1 May 1995
SUPERSEDING
MIL-S-19500/374B
3 May 1993

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER,
TYPES 2N3996 THROUGH 2N3999,
JAN, JANTX, JANTXV, JANS, JANHC AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN silicon, power transistors for use in high-speed power switching applications. Four levels of product assurance are provided for each device type as specified in MIL-S-19500. Two levels of product assurance are provided for the unencapsulated die.

1.2 Physical dimensions. See 3.3 and figure 1 (types 2N3996 and 2N3997, 4 lead stud package), figure 2 (types 2N3998 and 2N3999, 3 lead stud package), figure 3, and figure 4 (JANHC and JANKC).

1.3 Maximum ratings.

| $P_T$ 1/ $T_A$ = +25°C | $P_T$ 2/ $T_C$ = +100°C | $V_{EBO}$ | $V_{CBO}$ | $V_{CEO}$ | $I_B$ | $I_C$ | $I_C$ 3/ | $T_{STG}$ and $T_J$ | $R_{\theta JC}$ |
|---|---|---|---|---|---|---|---|---|---|
| W | W | V dc | V dc | V dc | A dc | A dc | A dc | °C | °C/W |
| 2 | 30 | 8 | 100 | 80 | 0.5 | 5.0 | 10 | -65 to +200 | 3.33 |

1/ Derate linearly, 11.4 mW/°C for $T_A \geq +25°C$.
2/ Derate linearly, 300 mW/°C for $T_C \geq +100°C$.
3/ This value applies for $t_p \leq 1$ ms, duty cycle ≤ 50%.

1.4 Primary electrical characteristics at $T_C$ = +25°C.

| Limit | $h_{FE2}$ 1/ $V_{CE}$ = 2 V dc; $I_C$ = 1 A dc | | $\lvert h_{fe} \rvert$ $V_{CE}$ = 5 V dc; $I_C$ = 1 A dc; f = 10 MHz | $V_{BE(sat)2}$ 1/ $I_C$ = 5 A dc; $I_B$ = 500 mA dc | $V_{CE(sat)2}$ 1/ $I_C$ = 5 A dc; $I_B$ = 500 mA dc | $C_{obo}$ $V_{CB}$ = 10 V dc $I_E$ = 0 100 kHz ≤ f ≤ 1 MHz |
|---|---|---|---|---|---|---|
| | 2N3996 2N3998 | 2N3997 2N3999 | | V dc | V dc | pF |
| Minimum | 40 | 80 | 3 | ---- | ---- | --- |
| Maximum | 120 | 240 | 12 | 1.6 | 2.0 | 150 |

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Projects Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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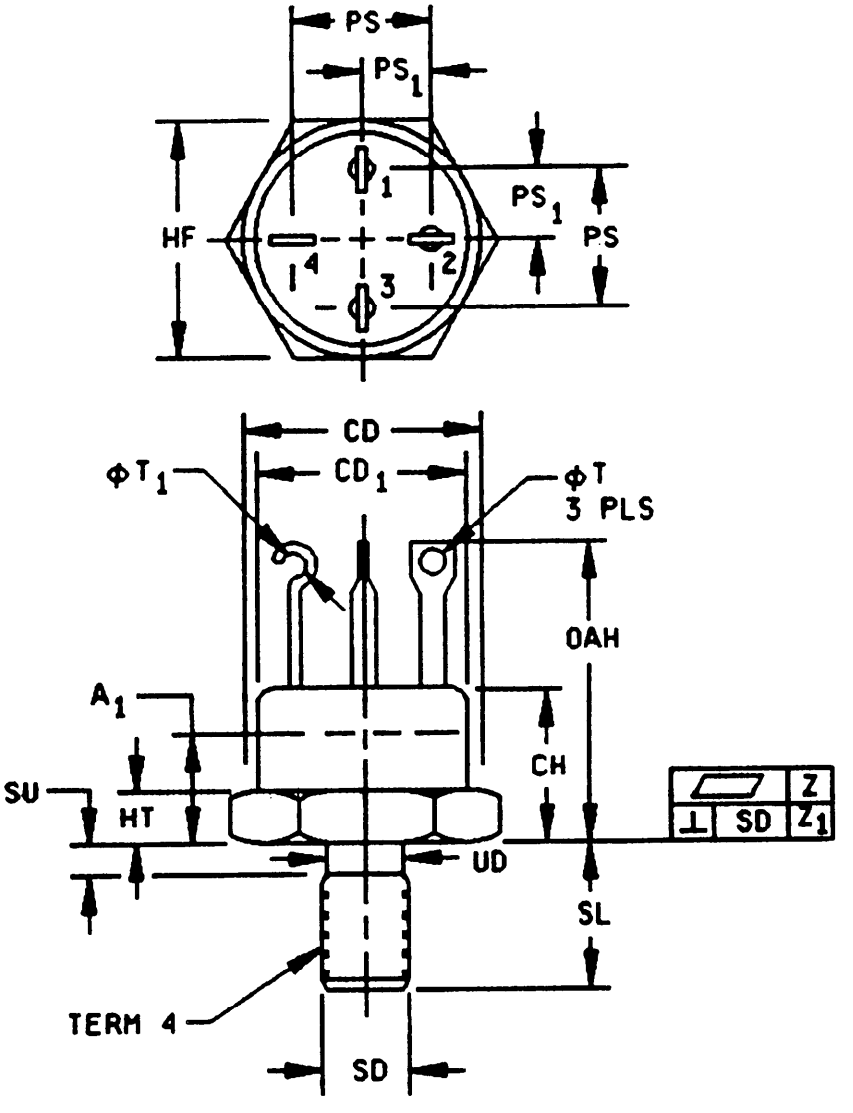



FIGURE 1. Physical dimensions for transistor types 2N3996 and 2N3997.



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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CH | .345 | .400 | 8.76 | 10.16 | |
| $A_1$ | | .250 | | 6.35 | 3 |
| CD | .318 | .380 | 8.08 | 9.65 | |
| $CD_1$ | .370 | .437 | 9.40 | 11.10 | 3 |
| HF | .424 | .437 | 10.77 | 11.10 | |
| PS | .180 | .215 | 4.57 | 5.46 | 5 |
| $PS_1$ | .080 | .110 | 2.03 | 2.79 | 5 |
| HT | .090 | .140 | 2.29 | 3.56 | 2,6 |
| OAH | .575 | .675 | 14.61 | 17.15 | 1 |
| UD | .155 | .189 | 3.94 | 4.80 | |
| SL | .400 | .455 | 10.16 | 11.56 | |
| SU | | .078 | | 1.98 | 7 |
| øT | .040 | .065 | 1.02 | 1.65 | |
| $øT_1$ | .040 | .065 | 1.02 | 1.65 | 4 |
| SD | .190-32 UNF-2A | | | | 8 |
| Z | | .002 | | 0.05 | |
| $Z_1$ | | .006 | | 0.15 | |

NOTES:
1. Terminal 1, emitter; terminal 2, base; terminal 3, collector; terminal 4, case.
2. Chamfer or undercut on one or both ends of hexagonal portion is optional.
3. The outline contour with the exception of the hexagon is optional within cylinder defined by $CD_1$ and $A_1$.
4. Terminal r can be flattened and pierced or hook type. A visual index is required when the flattened and pierced tab terminal contour (identical to the adjacent terminals) option is used. The case terminal (hook) is mechanically connected to the case. The other three terminals shall be electrically isolated from the case.
5. Angular orientation of terminals with respect to hexagon is optional.
6. HT dimension does not include sealing flanges.
7. SU is the length of incomplete or undercut threads.
8. SD is the pitch diameter of coated threads. Reference: Screw threads standards for Federal Service Handbook H28, part I.
9. Dimensions are in inches.
10. Metric equivalents are giving for general information only and are based upon 1.00 inch = 25.4 mm.
11. In accordance with ANSI Y14.5M, diameters are equivalent to øx symbology.

FIGURE 1. Physical dimensions for transistor types 2N3996 and 2N3997 - Continued.



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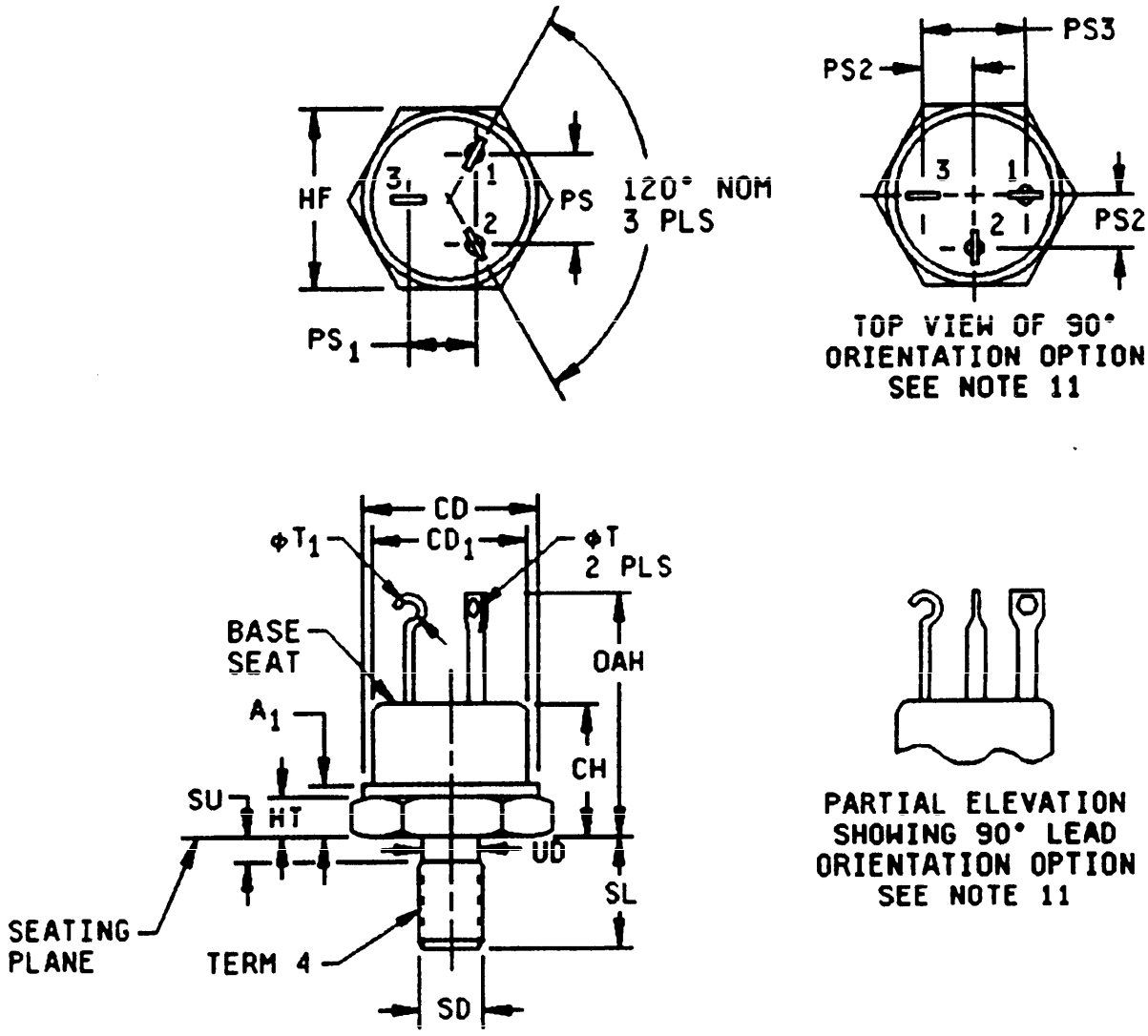



FIGURE 2. Physical dimensions for transistor types 2N3998 and 2N3999.



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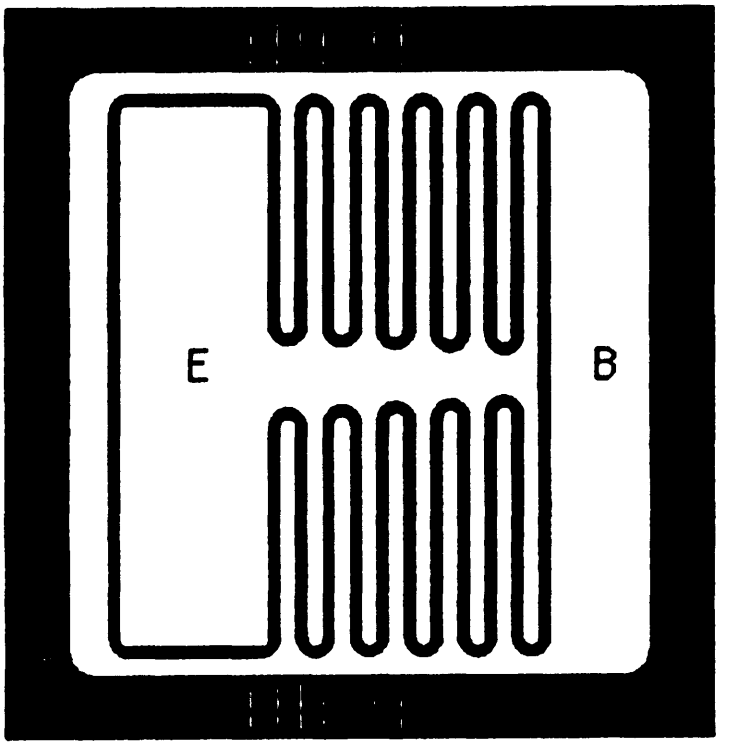
| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| CH | .345 | .400 | 8.76 | 10.16 | |
| $A_1$ | | .250 | | 6.35 | |
| $CD_1$ | .318 | .380 | 8.08 | 9.65 | |
| CD | .370 | .437 | 9.40 | 11.10 | |
| HF | .424 | .437 | 10.77 | 11.10 | |
| PS | .125 | .165 | 3.18 | 4.19 | 4,7,8 |
| $PS_1$ | .110 | .145 | 2.79 | 3.68 | 4,7 |
| $PS_2$ | .090 | .140 | 2.29 | 3.56 | 4,7,8 |
| $PS_3$ | .185 | .215 | 4.70 | 5.46 | 4,7,8 |
| HT | .090 | .140 | 2.29 | 3.56 | |
| OAH | .575 | .675 | 14.61 | 17.15 | 5 |
| UD | .155 | .189 | 3.94 | 4.80 | |
| SL | .400 | .455 | 10.16 | 11.56 | |
| SU | | .078 | | 1.98 | 9 |
| ϕT | .040 | .065 | 1.02 | 1.65 | |
| $\phi T_1$ | .040 | .065 | 1.02 | 1.65 | |
| SD | .190-32 UNF-2A | | | | 3 |

NOTES:
1. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
2. Collector shall be electrically connected to the case. This terminal may be flattened and pierced only when the 90° option is used.
3. SD is the pitch diameter of coated threads. Reference: Screw thread standards for Federal Service Handbook H28, part I.
4. The orientation of the terminals in relation to the hex flats is not controlled.
5. All three terminals.
6. The case temperature may be measured anywhere on the seating plane within .125 (3.18 mm) of the stud.
7. Terminal spacing measured at the base seat only.
8. Dimensions PS, $PS_1$, $PS_2$, and $PS_3$ are measured from the centerline of terminals.
9. Maximum unthreaded dimension.
10. This dimension applies to the location of the center line of the terminals.
11. A 90° angle lead orientation as shown may be used at the option of the manufacturer. All dimensions of the basic outline except PS, $PS_1$, and the 120° lead angle apply to this option.
12. Terminal 1, emitter; terminal 2, base; terminal 3, collector.
13. A slight chamfer or undercut on one or both ends of the hexagonal is optional.
14. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions for transistor types 2N3998 and 2N3999 - Continued.



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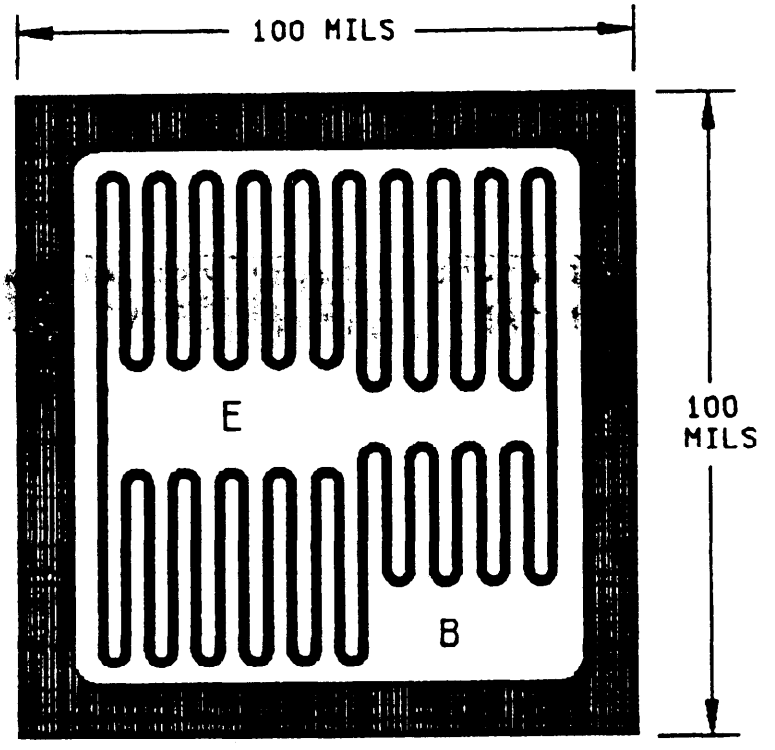

NOTES:

| | | |
|---|---|---|
| 1. | Chip size . . . . . | 82 X 82 mils |
| 2. | Chip thickness . . . . . | 6 to 12 mils |
| 3. | Top metal . . . . . | Aluminum 25,000 Å minimum, 30,000 Å nominal |
| 4. | Back metal . . . . . | Gold 2,500 Å minimum, 3,000 Å nominal |
| 5. | Backside . . . . . | Collector |
| 6. | Bonding pad . . . . . | B = 8 x 60 mils, E = 8 x 50 mils |

FIGURE 3. JANHC and JANKC (A-versions) die dimensions.



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NOTES:

| | | |
|---|---|---|
| 1. | Chip size . . . . . | 100 X 100 mils |
| 2. | Chip thickness . . . . . | 6 to 12 mils |
| 3. | Top metal . . . . . | Aluminum 25,000 Å minimum, 33,000 Å nominal |
| 4. | Back metal . . . . . | Gold 1,500 Å minimum, 2,500 Å nominal |
| 5. | Backside . . . . . | Collector |
| 6. | Bonding pad . . . . . | 12 x 30 mils |

FIGURE 4. JANHC and JANKC (B-versions) die dimensions.



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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500, and as follows.

3.3 Design, construction, and physical dimensions. Semiconductor devices shall be of the design, construction, and physical dimensions as specified in MIL-S-19500 and on figure 1, 2, 3, and 4 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein.

3.4 Marking. Devices shall be marked in accordance with MIL-S-19500. At the option of the manufacturer, the marking of the country of origin may be omitted from the body of the transistor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2.2 JANHC and JANKC devices. Qualification for JANHC and JANKC devices shall be in accordance with appendix H of MIL-S-19500. This testing may be performed utilizing a TO-5 package in lieu of the TO-59 or the TO-11.



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4.3 Screening (JANS, JANTXV, JANTX levels only). Screening shall be in accordance with MIL-S-19500 (table II) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table II of MIL-S-19500) | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| 3c 1/ | Thermal impedance (see 4.5.5) | Thermal impedance (see 4.5.5) |
| 9 | $I_{CES1}$ and $h_{FE2}$ | $I_{CES1}$ |
| 11 | $I_{CES1}$ and $h_{FE2}$; $\Delta I_{CES1}$ = 100% of initial value or 100 nA dc, whichever is greater; $\Delta h_{FE2}$ = +15%, -10% | $I_{CES1}$ and $h_{FE2}$; $\Delta I_{CES1}$ = 100% of initial value or 100 nA dc; whichever is greater. |
| 12 | See 4.3.1 | See 4.3.1 |
| 13a | Subgroups 2 and 3 of table I herein; $\Delta I_{CES1}$ = 100% of initial value or 100 nA dc, whichever is greater; $\Delta h_{FE2}$ = +15%, -10% | Subgroup 2 of table I herein, $\Delta I_{CES1}$ = 100% of initial value or 200 nA dc, whichever is greater; $\Delta h_{FE2}$ = +20%, -10% |
| 13b | Insulation resistance (terminal to case) Method 1016 of MIL-STD-750 (types 2N3996 and 2N3997 only); test condition B (short collector, emitter, and base terminals together); $R_{iso}$ = $10^9$ ohms minimum. | Insulation resistance (terminal to case) Method 1016 of MIL-STD-750 (types 2N3996 and 2N3997 only); test condition B (short collector, emitter, and base terminals together); $R_{iso}$ = $10^9$ ohms minimum. |

1/ May be performed anytime before screen 9.

4.3.1 Power burn-in conditions. Power burn-in conditions for all levels are as follows:

$V_{CE}$ = 25 V dc, ±5 V dc; $T_J$ = 187.5°C, ± 12.5°C; $T_A \leq$ 100°C.

4.3.2 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with appendix H of MIL-S-19500. Test limits and conditions shall be chosen by the supplier to demonstrate compliance with electrical characteristics specified in Table I herein.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500 and as specified herein. Group A inspection shall be performed on each sublot.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

| | | |
|---|---|---|
| a. | $I_M$ measure current | 10 mA |
| b. | $I_H$ forward heating current | 1.0 A to 3.0 A |
| c. | $t_H$ heating time | 100 ms |
| d. | $t_{MS}$ measurement delay time | 70 µs |
| e. | $V_H$ forward heating voltage | 10 V minimum |

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$(max) = 3.3°C/W.



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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX and JANTXV), of MIL-S-19500 and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B4 | 1037 | $V_{CE}$ = 20 V dc ±1.0 V dc; $P_T$ = 2 W minimum at $T_A$ = 25 ±3°C; $t_{on}$ = $t_{off}$ = 3 minutes minimum for 2000 cycles. No heat sink nor forced air cooling on the devices shall be permitted. |
| B5 | 1027 | $V_{CE}$ = 20 V dc ±1.0 V dc; 20°C ≤ $T_A$ ≤ 150°C for 96 hours. $P_T$ = 2 W at $T_A$ = 100°C or adjusted as required by the chosen $T_A$ to give an average lot $T_J$ = 275°C. |
| B7 | 3053 | Load condition C (unclamped inductive) (see figure 6), $T_A$ = 25°C ±3°C, duty cycle ≤ 10%; $R_S$ = 0.1Ω.<br><br>TEST 1 - $t_p$ = 640µS; $R_{BB1}$ 39Ω; $V_{BB1}$ = 20 V dc; $R_{BB2}$ = ∞; $V_{BB2}$ = 0; $V_{CC}$ = 10 V dc; $I_C$ = 4.3 A dc; L = 1 mH (0.5Ω, 5A) (Tower #7870 or equivalent).<br><br>TEST 2 - $t_p$ = 2.88 mS; $R_{BB1}$ 120Ω; $V_{BB1}$ = 20 V dc; $R_{BB2}$ = ∞; $V_{BB2}$ = 0; $V_{CC}$ = 10 V dc; $I_C$ = 1.4 A dc; L = 10 mH (0.11Ω, 12.5A) (Stancor C-2688 or equivalent) |
| B7 | 3053 | Safe operating area (switching) (destructive); Load condition B (clamped inductive) $T_A$ = 25°C; $I_B$ = 0.5 A dc, $I_C$ = 5 A dc; $V_{CC}$ = 55 V dc, L = 1.0 mH, $V_{BB2}$ = 3.0 V, $R_{BB2}$ = 20 Ω. |

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1037 | $V_{CE}$ = 20 V dc ±1.0 V dc; $P_T$ = 2 W minimum at $T_A$ = 25 ±3°C; $t_{on}$ = $t_{off}$ = 3 minutes minimum for 2000 cycles. No heat sink nor forced air cooling on the devices shall be permitted. |
| B5 | ---- | Not applicable. |
| B7 | 3053 | Load condition C (unclamped inductive) (see figure 6), $T_A$ = 25°C, duty cycle ≤ 10%; $R_S$ = 0.1Ω.<br><br>TEST 1 - $t_p$ = 640µS; $R_{BB1}$ 39Ω; $V_{BB1}$ = 20 V dc; $R_{BB2}$ = ∞; $V_{BB2}$ = 0; $V_{CC}$ = 10 V dc; $I_C$ = 4.3 A dc; L = 1 mH (0.5Ω, 5A) (Tower #7870 or equivalent).<br><br>TEST 2 - $t_p$ = 2.88 mS; $R_{BB1}$ 120Ω; $V_{BB1}$ = 20 V dc; $R_{BB2}$ = ∞; $V_{BB2}$ = 0; $V_{CC}$ = 10 V dc; $I_C$ = 1.4 A dc; L = 10 mH (0.11Ω, 12.5A) (Stancor C-2688 or equivalent) |
| B7 | 3051 | Safe operating area (switching) (destructive); Load condition B (clamped inductive) $T_A$ = 25°C; $I_B$ = 0.5 A dc, $I_C$ = 5 A dc; $V_{CC}$ = 55 V dc, L = 1.0 mH, $V_{BB2}$ = 3.0 V, $R_{BB2}$ = 20 Ω. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition A; weight = 7 lbs ±5 oz, application time = 15 s, tubulated leads only.<br>Test condition D2; Torque = 15 in-lb; application time = 15 s.<br>Test condition D1; torque = 8 in-oz; application time = 15 s; tubulated leads only. |
| C6 | 1026 | $V_{CE}$ = 20 V dc ±1 v dc; $P_T$ = 30 W at $T_C$ = 100°C. |



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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Case-temperature control for $h_{fe}$ test. To maintain the case temperature at less than +40°C for this test, the specified dc collector current should be applied for not longer than 10 seconds without employing a heat sink.

4.5.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be conducted as a case temperature ($T_C$) of +25°C.

4.5.4 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 0.833 A dc.
b. Collector to emitter voltage magnitude shall be 20 V dc.
c. Reference temperature measuring point shall be the case.
d. Reference point temperature shall be 25°C ≤ $T_R$ ≤ 75°C and recorded before the test is started.
e. Mounting arrangement shall be with heat sink to case.
f. Maximum limit of $R_{\theta JC}$ shall be 3.33°C/W.

4.5.5 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $\Delta V_{BE}$ measurements shall be performed in accordance with MIL-STD-750, method 3131. The maximum limit and conditions for $\Delta V_{BE}$ in screening (see table II of MIL-S-19500) shall be derived by each vendor by means of process control of actual measurements which characterizes the die attach process. When three lot date codes have exhibited control, the data from these three lots will be used to establish a fixed screening limit (not to exceed the group A limit). Once a fixed limit has been established, monitor all future sealing lots using a sample from each lot to be plotted on the applicable X and R chart.

4.5.5.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750. Method 3131 (read and record data $Z_{\theta JX}$) derived conditions limits and thermal response curve shall be supplied to the qualifying activity on the qualification lot prior to qualification approval.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance | 3101 | See paragraph 4.5.5 | $Z_{\theta JX}$ | | 3.3 | °C/W |
| Breakdown voltage, collector to base | 3001 | Bias condition D, $I_C$ = 10 µA dc, $I_B$ = 0 | $V_{(BR)CBO}$ | 100 | | V dc |
| Breakdown voltage, collector to emitter | 3011 | Bias condition D, $I_C$ = 50 mA dc, pulsed (see 4.5.1) | $V_{(BR)CEO}$ | 80 | | V dc |
| Collector to emitter cutoff current | 3041 | Bias condition C, $V_{CE}$ = 80 V dc, $V_{BE}$ = 0 | $I_{CES1}$ | | 200 | nA dc |
| Collector to emitter cutoff current | 3041 | Bias condition D, $V_{CE}$ = 60 V dc, $I_B$ = 0 | $I_{CEO}$ | | 10 | µA dc |
| Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = 5 V dc, $I_C$ = 0 | $I_{EBO1}$ | | 200 | nA dc |
| Emitter to base cutoff current | 3061 | Bias condition D, $V_{EB}$ = 8 V dc, $I_C$ = 0 | $I_{EBO2}$ | | 10 | µA dc |
| Base-emitter voltage (saturated) | 3066 | Test condition A, $I_C$ = 1 A dc; $I_B$ = 0.1 A dc, pulsed (see 4.5.1) | $V_{BE(sat)1}$ | 0.6 | 1.2 | V dc |
| Base-emitter voltage (saturated) | 3066 | Test condition A, $I_C$ = 5 A dc; $I_B$ = 0.5 A dc, pulsed (see 4.5.1) | $V_{BE(sat)2}$ | | 1.6 | V dc |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 1 A dc; $I_B$ = 0.1 A dc, pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | 0.25 | V dc |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = 5 A dc; $I_B$ = 0.5 A dc, pulsed (see 4.5.1) | $V_{CE(sat)2}$ | | 2 | V dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2 V dc; $I_C$ = 50 mA dc, pulsed (see 4.5.1) | $h_{FE1}$ | | | |
| 2N3996, 2N3998 | | | | 30 | | |
| 2N3997, 2N3999 | | | | 60 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2 V dc; $I_C$ = 1 A dc, pulsed (see 4.5.1) | $h_{FE2}$ | | | |
| 2N3996, 2N3998 | | | | 40 | 120 | |
| 2N3997, 2N3999 | | | | 80 | 240 | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 5 V dc; $I_C$ = 5 A dc, pulsed (see 4.5.1) | $h_{FE3}$ | | | |
| 2N3996, 2N3998 | | | | 15 | | |
| 2N3997, 2N3999 | | | | 20 | | |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 | | | | | | |
| High temperature operation | | $T_C$ = +150°C | | | | |
| Collector to emitter cutoff current | 3041 | Bias Condition C, $V_{CE}$ = 80 V dc, $V_{BE}$ = 0 | $I_{CES2}$ | | 50 | µA dc |
| Low-temperature operation | | $T_C$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 2 V dc; $I_C$ = 1 A dc, pulsed (see 4.5.1) | $h_{FE4}$ | | | |
| 2N3996, 2N3998 | | | | 10 | | |
| 2N3997, 2N3999 | | | | 20 | | |
| Subgroup 4 | | | | | | |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 10 V dc; $I_E$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | | 150 | pF |
| Magnitude of common emitter small-signal short-circuit forward current transfer ratio | 3306 | $V_{CE}$ = 5 V dc; $I_C$ = 1 A dc; f = 10 MHz, (see 4.5.2) | $h_{fe}$ | 3 | 12 | |
| Switching parameters: | | | | | | |
| 2N3996 and 2N3998 | | | | | | |
| pulse delay time | | See figure 5 | $t_d$ | | 100 | ns |
| pulse rise time | | See figure 5 | $t_r$ | | 240 | ns |
| pulse storage time | | See figure 5 | $t_s$ | | 1.4 | µs |
| pulse fall time | | See figure 5 | $t_f$ | | .3 | µs |
| $t_{on}$ | | $t_d + t_r$ | $t_{on}$ | | .3 | µs |
| $t_{off}$ | | $t_s + t_f$ | $t_{off}$ | | 1.5 | µs |
| 2N3997 and 2N3999 | | | | | | |
| pulse delay time | | See figure 5 | $t_d$ | | 100 | ns |
| pulse rise time | | See figure 5 | $t_r$ | | 240 | ns |
| pulse storage time | | See figure 5 | $t_s$ | | 1.75 | µs |
| pulse fall time | | See figure 5 | $t_f$ | | .3 | µs |
| $t_{on}$ | | $t_d + t_r$ | $t_{on}$ | | .3 | µs |
| $t_{off}$ | | $t_s + t_f$ | $t_{off}$ | | 2.0 | µs |
| Subgroup 5 | | | | | | |
| Safe operating area (DC) | 3051 | $T_C$ = 100°C; power application time = 1.0 sec. (see figure 6) | | | | |
| Test 1 | | $V_{CE}$ = 80 V dc; $I_C$ = .080 A dc | | | | |
| Test 2 | | $V_{CE}$ = 20 V dc; $I_C$ = 1.5 A dc | | | | |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 5 - Con't | | | | | | |
| Safe operating area (clamped inductive) | | (See figures 8 and 9); (clamped inductive load) $T_A$ = 25°C; $I_B$ = 0.5 A dc; $I_C$ = 5 A dc; $V_{CC}$ = 15 V dc; Load condition C | | | | |
| Electrical measurements | | See table II, steps 1, 4, 5, and 8. | | | | |
| Subgroup 6 and 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-S-19500.



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TABLE II. Groups A, B, and C electrical measurements.

| Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1 | Collector to emitter cutoff current | 3041 | Bias condition C; $V_{CE}$ = 80 V dc $V_{BE}$ = 0 | $I_{CES1}$ | | 200 | nA dc |
| 2 | Collector to emitter cutoff current | 3041 | Bias condition C; $V_{CB}$ = 80 V dc $V_{BE}$ = 0 | $I_{CES1}$ | | 400 | nA dc |
| 3 | Collector to emitter cutoff current | 3041 | Bias condition D; $V_{CB}$ = 60 V dc $V_{BE}$ = 0 | $I_{CEO}$ | | 10 | µA dc |
| 4 | Emitter to base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 5 V dc $I_C$ = 0 | $I_{EBO1}$ | | 200 | nA dc |
| 5 | Breakdown voltage collector to emitter | 3011 | Bias condition D; $I_C$ = 50 mA dc; $I_B$ = 0 pulsed (see 4.5.1) | $V_{(BR)CEO}$ | 80 | | V dc |
| 6 | Base emitter voltage (saturated) | 3066 | Test condition A; $I_C$ = 1 A dc $I_B$ = 0.1 A dc pulsed (see 4.5.1) | $V_{BE(SAT)1}$ | 0.6 | 1.2 | V dc |
| 7 | Collector to emitter voltage (saturated) | 3071 | $I_C$ = 1.0 A dc $I_B$ = 0.1 A dc pulsed (see 4.5.1) | $V_{CE(sat)1}$ | | 0.25 | V dc |
| 8 | Forward-current transfer ratio | 3076 | $V_{CE}$ = 2 V dc $I_C$ = 1 A dc pulsed (see 4.5.1) | $h_{FE2}$ | | | |
| | 2N3996, 2N3998 | | | | 40 | 120 | |
| | 2N3997, 2N3999 | | | | 80 | 240 | |
| 9 | Forward-current transfer ratio | 3076 | $V_{CE}$ = 2 V dc $I_C$ = 1 A dc pulsed (see 4.5.1) | $\Delta h_{FE2}$ 1/ | | ±20% change from initial reading. | |
| 10 | Collector to emitter cutoff current | 3041 | Bias condition D; $V_{CE}$ = 60 V dc $I_B$ = 0 | $\Delta I_{CEO}$ 1/ | | 100% of initial value or 100 nA dc, whichever is greater. | |
| 11 | Collector to emitter voltage (saturated) | 3071 | $I_C$ = 1 A dc $I_B$ = 0.1 A dc pulsed (see 4.5.1) | $\Delta V_{CE(SAT)1}$ 1/ | | ±50% mV dc change form previously measured value. | |
| 12 | Thermal impedance | 3131 | See 4.4.1 & 4.5.5 | $Z_{\theta JX}$ | | 3.3 | °C/W |

See footnotes at end of table.



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1/ The devices which exceed the group A limits for this test shall not be acceptable.

2/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 3, 4, 6, 7, and 8.

b. Subgroups 4 and 5, see table II herein, steps 1, 3, 4, 6, 7, 8, 9, 10, and 11.

c. Subgroup 7, see table II herein, steps 1, 3, 7, and 9.

3/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, 4, 7, and 8.

b. Subgroup 3, see table II herein, steps 2, 5, 7, and 9.

c. Subgroup 6, see table II herein, steps 2, 5, 7, 8, and 10.

d. Subgroup 7, see table II herein, steps 1, 3, 7, and 9.

4/ The electrical measurements for table V of MIL-S-19500 are as follows:

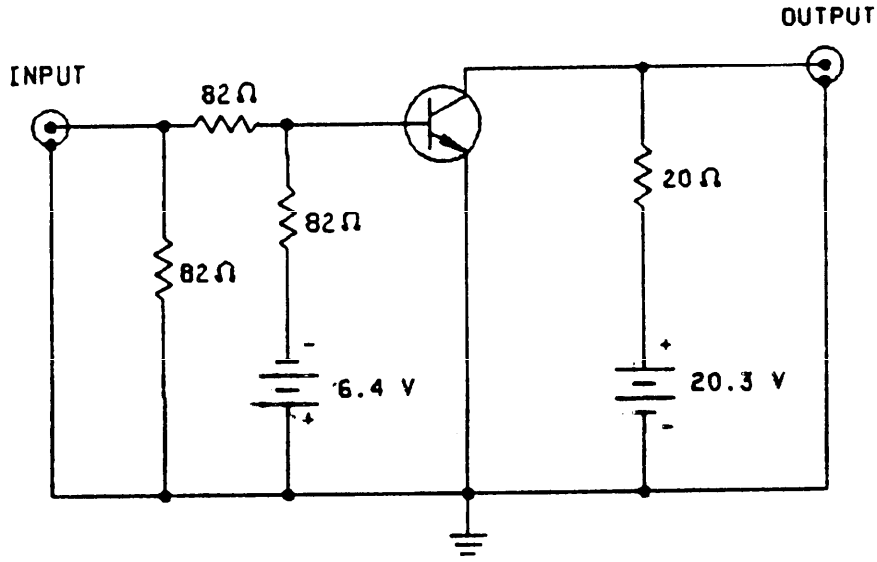
a. Subgroup 2, see table II herein, steps 1, 4, 6, and 11 for JANS and steps 1, 4, 5, 6, and 8 for JANTX and JANTXV.

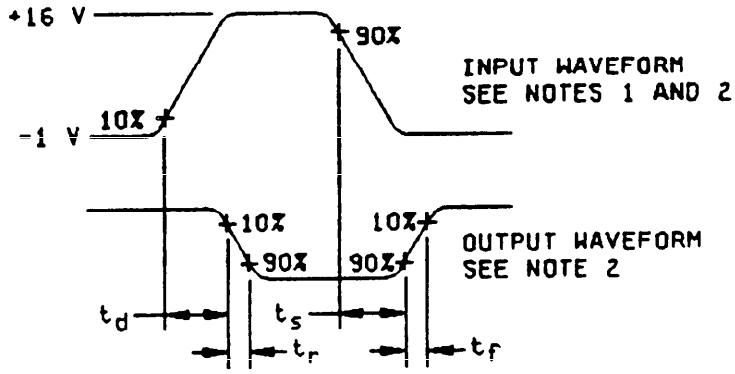
b. Subgroup 3, see table II herein, steps 5 and 6 for all levels.

c. Subgroup 6, see table II herein, steps 3, 4, 6, 7, 9, and 10 for JANS and steps 2, 7, and 9 for JANTX and JANTXV.



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NOTES:
1. The input waveform is supplied by a generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, $Z_{out} = 50\ \Omega$, pw = 2 µs, duty cycle ≤ 2 percent.
2. Waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 15$ ns, $R_{in} \geq 10$ MΩ, $C_{in} \leq 11.5$ pF.
3. Resistors must be non-inductive types.
4. The d-c power supplies may require additional by-passing in order to minimize ringing.

FIGURE 5. Pulse response test circuit.



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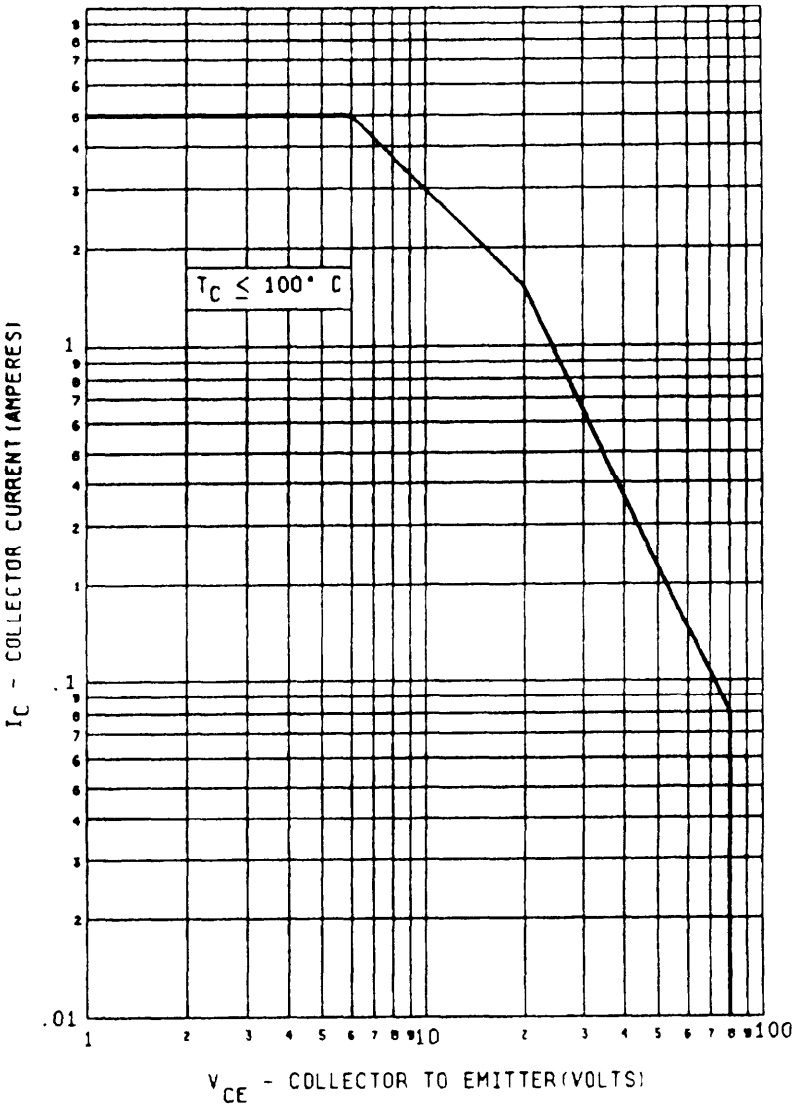



FIGURE 6. Maximum safe operating area graph (dc).



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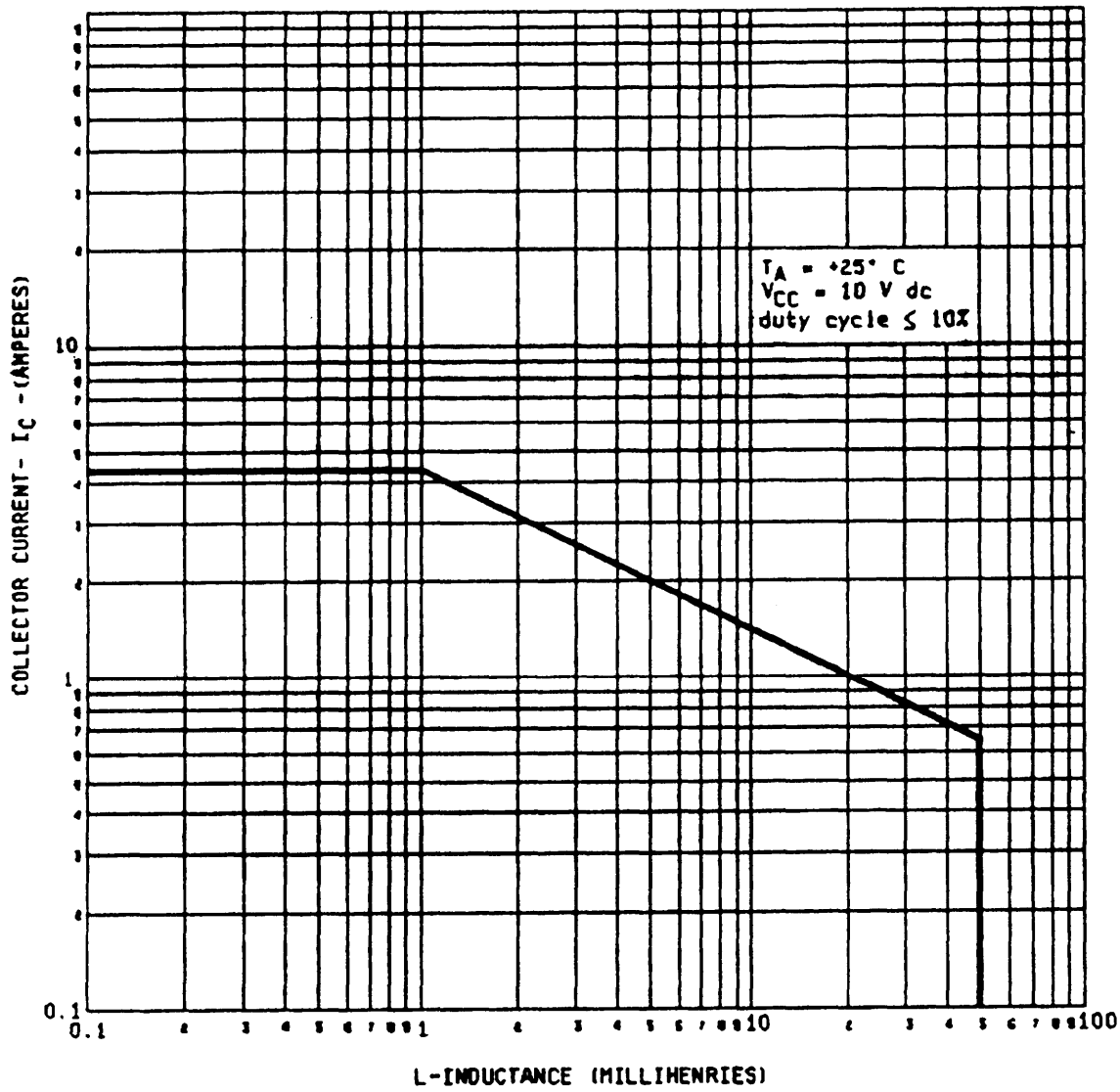
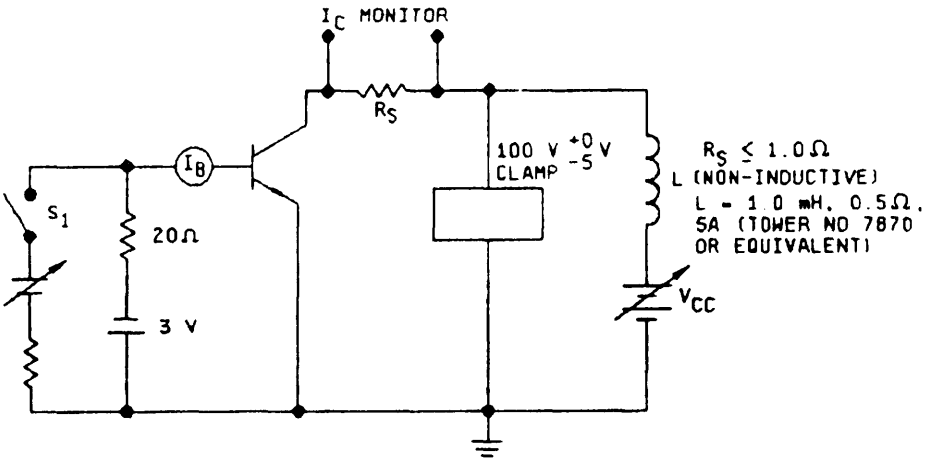



FIGURE 7. Safe operating area for switching between saturation and cutoff - unclamped inductive load.



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Procedure:
1. With switch $S_1$ closed, set the specified test conditions.
2. Open $S_1$. Device fails if clamp voltage not reached.
3. Perform specified end point test.

FIGURE 8. Clamped inductive sweep test circuit.

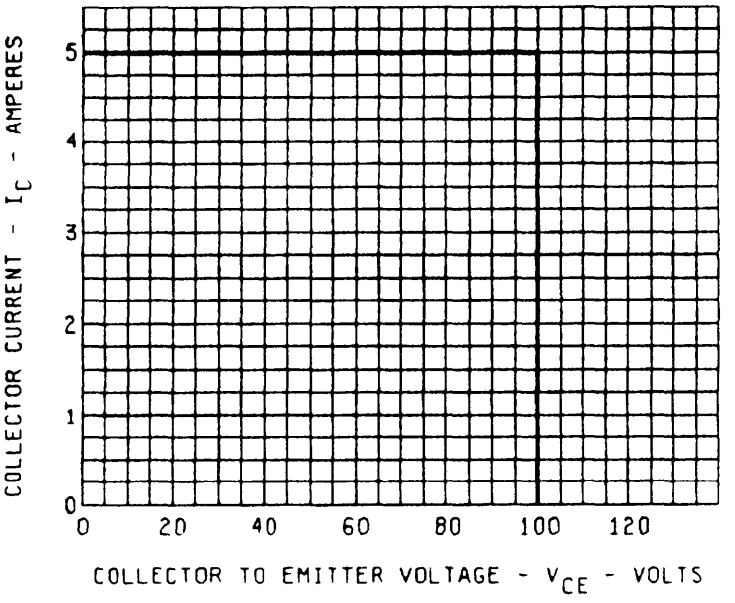



FIGURE 9. Safe operating area for switching between saturation and cutoff - clamped inductive load.



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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Product assurance level and type designation.

6.3 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N3996) will be identified on the QPL.

| JANHC and JANKC ordering information | | |
|---|---|---|
| | Manufacturer 1/ | |
| PIN | 33178 | 34156 |
| 2N3996 | JANHCA2N3996 | JANHCB2N3996 |
| 2N3997 | JANHCA2N3997 | JANHCB2N3997 |
| 2N3998 | JANHCA2N3998 | JANHCB2N3998 |
| 2N3999 | JANHCA2N3999 | JANHCB2N3999 |

1/ For JANKC level, replace the JANHC prefix with JANKC.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AV
Air Force - 85, 99
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5961-1693)



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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-S-19500/374C | 2. DOCUMENT DATE (YYMMDD)<br>95/05/01 |
|---|---|---|

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER, TYPES 2N3996 THROUGH 2N3999, JAN, JANTX, JANTXV, JANS, JANHC AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME<br>Greg Rose | b. TELEPHONE (Include Area Code)<br>(1) Commercial<br>301-731-8694<br>(2) AUTOVON |
|---|---|
| c. ADDRESS (Include Zip Code)<br>NASA/Parts Projects Office (NPPO),<br>NASA Goddard Space Flight Center<br>Code 310.A<br>Greenbelt, MD 20771 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289 2340 |



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